

EE491–EE492

Senior Design Project Proposal

Multi-Sensor Cough Analysis and State Classification with Machine Learning

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1 Introduction

Respiratory illnesses such as asthma, pneumonia, chronic obstructive pulmonary disease (COPD), and COVID-19 affect millions of people worldwide. Early detection and continuous monitoring are very important for accurate diagnosis and treatment. Traditional tools like stethoscopes have limitations as they depend on the doctor’s listening skills and the patient’s own reports. With advances in biomedical signal processing and machine learning, it has become possible to extract valuable information from physiological signals and cough sounds [1], [2].

This project will use a lab-built device that integrates four sensors: an accelerometer, a strain sensor, an ambient microphone, and a pulmonary microphone. Similar sensor-based systems have already shown promising results in cough detection and monitoring [3]. The goal of this project is to analyze cough events in detail by determining the patient’s physical state (walking, sitting, running, or sleeping) during coughing and to detect whether the cough originates from the patient or from the surrounding environment.

2 Objectives

The main objectives of this project are:

1. To create a machine learning model that can distinguish between patient coughs and background coughs.
2. To identify the patient’s physical state during cough events using signals from an accelerometer, strain sensor, ambient microphone, and pulmonary microphone [3].

After getting satisfactory results from the model, I will compare and explore both classical machine learning and deep learning approaches [1], [4], compare single-sensor and multi-sensor performance, and if time permits, develop a basic real-time monitoring application.

3 Approach and Methodology

The project will begin with data collection and labeling using the four-sensor device. Recordings will be made while subjects perform activities such as sitting, walking, running, and resting. Cough events will be annotated manually and supported with automatic detection methods.

In the preprocessing stage, audio signals will be filtered and denoised, while accelerometer and strain data will be smoothed and normalized. Cough segments will be identified using short-time energy and spectral features [2]. All signals will then be divided into time windows to prepare for feature extraction.

For feature extraction, audio features such as mel-frequency cepstral coefficients (MFCCs), spectrograms, and spectral entropy will be used. From motion and strain data, statistical features and respiratory cycle patterns will be calculated [3].

For classification, several models will be trained, including Support Vector Machines (SVM), Random Forest, Gradient Boosting, and feedforward neural networks. Deep learning models like Convolutional Neural Networks (CNNs) or Long Short-Term Memory (LSTM) networks may also be tested, especially for spectrogram-based audio analysis [1], [4]. Sensor fusion methods will be applied to combine audio and motion data to improve overall accuracy.

Finally, the system will be evaluated using metrics such as accuracy, F1-score, sensitivity and specificity. Comparisons will be made between single-sensor and multi-sensor setups. To measure generalization, Cross-validation strategies such as k-fold validation and leave-one-subject-out testing will be used.

4 Expected Results and Accomplishments

The main expected outcome is a system that can:

- Correctly identify the patient’s current physical state during cough events.
- Distinguish between coughs coming from the patient and those coming from the environment.

By combining audio and motion data, the project aims to achieve reliable accuracy. The final accomplishment will consist of a complete processing pipeline that includes preprocessing, feature extraction, classification, and evaluation.

Performance will be judged based on three priorities:

1. Accurate detection of patient cough events with high sensitivity and robustness to noise.
2. Correct classification of physical activity during coughing.
3. Reliable distinction between patient coughs and background coughs.

If time allows, an additional achievement will be the development of a simple interface prototype for real-time use.

5 Realistic Constraints

The success of this project depends on several factors:

- Availability of high-quality and well-labeled data from the four sensors.

- Noise in recordings, which may reduce classification accuracy [3].
- Synchronization of multimodal signals, which is technically challenging.
- Computational resources needed for audio spectrograms, accelerometer data, and strain signals [1].
- The risk of overfitting when working with limited datasets [4].
- Ethical requirements, including secure handling of sensitive audio data.

a. Social, Environmental, and Economic Impact

This project can improve healthcare by providing objective cough and activity analysis, supporting early diagnosis and better treatment [2]. Socially, it can benefit patients by enabling remote monitoring and reducing unnecessary hospital visits. Environmentally, fewer hospital trips may help lower energy use and transportation emissions. Economically, the system could reduce costs for patients and healthcare providers by lowering the need for expensive diagnostic procedures [4].

b. Cost Analysis

The only cost of the project is labor. Assuming the student researcher works about 10 hours per week during the semester at an hourly rate of \$15, the labor cost is estimated at around \$2400 per semester and about \$4800 for two semesters. Since the device is already available in the lab, no additional hardware expenses are expected.

c. Standards

The system must follow ISO/IEEE 11073 standards for medical device communication and interoperability. On the software side, the development process should align with IEEE 12207 standards for software life cycle management. In addition, compliance with Turkish and EU Medical Device Regulations (MDR) is essential to ensure safety, privacy, and ethical use in healthcare.

6 Time Schedule

The following monthly plan is tentative and may be adjusted according to project progress and data availability.

- **Month 1:** Literature review, background study, familiarization with device and dataset.

- **Month 2:** Data acquisition (multi-sensor), preprocessing (filtering, normalization, synchronization).
- **Month 3:** Segmentation (sliding windows), feature extraction (MFCCs, statistical features).
- **Month 4–5:** Baseline ML model creation and evaluation (SVM, RF, GB), cross-validation.
- **Month 6–7:** Deep learning model creation and evaluation (CNN, LSTM), sensor fusion, hyperparameter tuning.
- **Month 8:** Comparison of models (ML vs DL), robustness testing, final report preparation and real-time prototype demo (if time permits).

References

- [1] J. Amoh and K. Odame, “Deep neural networks for identifying cough sounds,” *IEEE Trans. Biomed. Circuits Syst.*, vol. 10, no. 5, pp. 1003–1011, 2016.
- [2] Y. Shi, H. Liu, Y. Wang, M. Cai, and W. Xu, “Theory and application of audio-based assessment of cough,” *J. Sensors*, vol. 2018, Art. no. 9845321, 2018.
- [3] T. Otoshi, T. Nagano, S. Izumi, D. Hazama, *et al.*, “A novel automatic cough frequency monitoring system combining a triaxial accelerometer and a stretchable strain sensor,” *Sci. Rep.*, vol. 11, Art. no. 9973, 2021.
- [4] D. Sanchez-Morillo, D. Sales-Lerida, B. Priego-Torres, and A. León-Jiménez, “Cough detection using acceleration signals and deep learning techniques,” *Electronics*, vol. 13, no. 12, Art. no. 2410, 2024.